

and operational best practices. Start by optimising data layout store files in open formats like Parquet with sizes between 100-500MB to balance performance and metadata overhead. Use Apache Iceberg's hidden partitioning and schedule regular compaction jobs using Spark or Flink to mitigate the small file problem. Maintaining clean and consistent metadata is equally vital: enforce naming conventions, periodically prune unused snapshots, and consider using versioned catalogues like Project Nessie for better control over schema evolution and rollback.

On the performance and governance front, enable predicate pushdown and metadata pruning in Trino to accelerate SQL queries across Iceberg tables. For security, integrate access control solutions such as Apache Ranger or AWS Lake Formation to implement table- and column-level permissions. Data lineage and auditability can be enhanced using tools like Marquez or OpenLineage. Finally, monitor system health through Trino's query metrics and Iceberg's snapshot lifecycle, setting alerts for ingestion failures or metadata bloat. These practices ensure your lakehouse remains production-ready, cost-efficient, and aligned with enterprise-grade analytics and compliance needs.

The integration of Apache Iceberg and Trino allows for a robust and adaptable method of constructing new data lakehouse architecture. This architecture combines data lakes' scalability and cost-effectiveness with data warehouses' transactional integrity and query performance while open standards are embraced. By isolating storage, metadata and compute, and federating SQL access across heterogeneous sources, organisations can enable real-time analytics, BI, and machine learning from a single platform. As data continues to get denser and more complex, the Iceberg-Trino stack gives us a future-proof base to deliver governed, high-performance insights at scale. **END** 🐼

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🔗 By: Dulari Bhatt and Shivam Maurya

Dulari Bhatt is master trainer at Edunet Foundation and has around 12 years of academic experience. Her core expertise is in computer vision, machine learning, deep learning, and data analytics.

Shivam Maurya is an open source enthusiast and a researcher.

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